Giorgio Giannone

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I am a Senior Research Scientist with the AI Foundations Team at **Waymo** and Research Affiliate with the DeCoDE Lab at **MIT**.

My research lies at the intersection of Generative Modeling and Probabilistic Inference, with a focus on Efficient Methods for Data-constrained Domains in Engineering and Science.

My current interest areas include Inference-Time Scaling, Test-Time Adaptation, Vision-Language Alignment, and Few-Shot Generation.

Experience

Senior Research Scientist, Waymo — Mountain View, California, USA
Oct 2026 - Present

- AI Foundations Team
  - Foundation Models for Simulation and Planning
  - Inference-Time Scaling for Autonomous Driving

Principal Research Scientist, Red Hat — Boston, Massachusetts, USA
June 2025 - Oct 2026

- AI Innovation and Harness Engineering Team
  - Probabilistic Inference for Vision and Language Models
  - Inference-Time Scaling and Reasoning for LLMs (ICML, NeurIPS)
  - Data Efficient RL for Foundation Models
  - `its-hub` Algorithm Development and Contributions
  - Context Optimization for Research-centric Harness Engineering

Research Affiliate, Massachusetts Institute of Technology — Cambridge, Massachusetts, USA
Jan 2026 - Present

- DeCoDE Lab. Department of Mechanical Engineering
  - Inference-Time Scaling for Constrained Generative Design
  - Iterative Self-Training for CAD Program Synthesis (ICML)

Applied Scientist, Amazon — Seattle, Washington, USA
April 2024 - June 2025

- Home Innovation and GenAI Team
  - Grounded Vision-Language Models
  - Evaluation for Text-to-Image Models (CVPR)
  - Detection and Ranking Algorithms for Amazon Visual Shopping
  - Subject-Driven Generative Models for AI Creative Studio

Visiting Researcher, UCL Centre for Artificial Intelligence — London, UK
Jan 2024 - March 2024

- Host: David Barber
  - Multi-Resolution Convolutional Models for Long Sequences (NeurIPS)
  - Bayesian Inference for Language Models

PhD Internships and Collaborations

Researcher (PhD Intern), Microsoft Research — Cambridge, MA, USA, *Jun 2023 - Sept 2023*

- ML and Statistics Group. Hosts: David Alvarez Melis, Nicolo Fusi
- Dynamic Vocabulary Augmentation for LLMs

Research Collaborator, MIT-IBM AI Lab — Cambridge, MA, USA, *Jan 2023 - June 2023*

- Model Alignment Team. Host: Akash Srivastava

- Generative Models for Systems with Constraints (NeurIPS)
- Aligning Language Models with Negative Data
- Specialized Language Models for Enterprise Domains

Research Scientist (PhD Intern), IBM Research — Zurich, Switzerland, *Jun 2022 - Nov 2022*

- Accelerated Discovery Team. Hosts: Matteo Manica, Teodoro Laino
- Multitask Language Models for Text and Chemistry (ICML)
- Open-source library GT4SD for conditional generative models

Applied Scientist (PhD Intern), Amazon Science — Cambridge & London, UK, *Jul 2021 - Oct 2021*

- Alexa Team. Hosts: Yunlong Jiao, Emine Yilmaz
- Domain Agnostic Subpopulation Generalisation

Research Engineer, NNAISENSE — Lugano, Switzerland, *Jan 2019 - Jan 2020*

- Deep Learning Team. Managers: Christian Osendorfer, Jonathan Masci
  - Structured Latent Variable Models
  - NeuralODE Algorithms for High-Range Event Camera Streams

Co-Founder, SecretAIry (formerly GAiA) — Rome, Italy, *July 2017 - Jan 2019*

- Chatbots to enhance Workplace Communication
  - Selected among 100+ startups to join the EnLabs Incubator

Education

PhD, Generative Machine Learning — Technical University of Denmark, Lyngby, Denmark, *June 2020 - Dec 2023*

- Few-Shot Generative Models (ICML)
- Multitask Language Models for Conditional Molecule Generation (ICML)
- Diffusion Models for Generative Engineering Design and Topology Optimization (NeurIPS)
- Thesis: Learning Generative Models with Limited Data
  - Supervisor: Ole Winther; Co-supervisor: Søren Hauberg

Visiting PhD Student, MIT School of Engineering — Cambridge, Massachusetts, USA, *Jan 2023 - Sept 2023*

- Constrained Diffusion Models for Engineering Design (NeurIPS & Patent)
- Improving Generative Constraint Satisfaction using Invalid Designs (TMLR)
- Evaluating Vision-Language Models for Engineering Tasks (Journal)
- Research on LLM Agents for CAD design. Co-developer of `text2cad`.
  - Host: Faez Ahmed, DeCoDE Lab

Master's Degree, Data Science — Sapienza University, Rome, Italy, *Sept 2016 - Nov 2018*

- Excellence Path & Summa Cum Laude
- Thesis: Multimodal Learning for Scene Understanding
  - Supervisor: Aris Anagnostopoulos; External Supervisor: Boris Chidlovskii

Visiting Graduate Student, NYU Tandon School of Engineering — NYC, New York, USA, *Sept 2017 - Jan 2018*

- Visualization and Data Analytics Research Center. Host: Enrico Bertini
  - Built an interactive entity retrieval tool to investigate 10M documents

Master's Degree, Mechanical Engineering — Sapienza University, Rome, Italy, *Sept 2014 - Jan 2017*

- Summa Cum Laude
- Thesis: Bubble Dynamics in Turbulent Shear Flows
  - Supervisor: Carlo Massimo Casciola; Co-supervisor: Paolo Gualtieri

Bachelor's Degree, Mechanical Engineering — Sapienza University, Rome, Italy, *Sept 2009 - May 2014*

- Thesis: Rapid Prototyping of Metallic Manufacturing

Selected Publications & Patents

DeliberateRL: Shaping the Practice Distribution for Compute-Efficient RLVR under-review
Sudalairaj, Xu, Srivastava, Giannone *2026*

Intrinsic Selection and Particle Resampling for ITS Beyond Domain Verifiability NeurIPS
Giannone, Eyceoz, Baig, Sudalairaj ,Doris, Ahmed, Srivastava, Xu *2026*

Bootstrapping Image-to-CAD Program Synthesis via Geometric Feedback ICML
Giannone, Doris, Nobari, Xu, Srivastava, Ahmed *2026*

Mitigating Premature Exploitation in Particle-based Monte Carlo for ITS ICML
Giannone, Xu, Nayak, Awhad, Sudalairaj, Xu, Srivastava *2026*

Generative optimization models for machine learning US Patent (MIT & IBM)
Giannone, Srivastava, Ahmed *2025*

Feedback-Driven Vision-Language Alignment via SVP TMLR
Giannone, Perevodchikov, Feng, Li, Chen, Martinez *2026*

Be More Specific: Evaluating Object-centric Realism in Synthetic Images CVPR
Liang, Corneanu, Feng, Giannone, Martinez *2025*

Evaluating Vision-Language Models for Engineering Design Springer Artificial Intelligence Review
Picard, Edwards, Doris, Mann, Giannone, Alam, Ahmed *2025*

NITO: Neural Implicit Fields for Resolution-free Topology Optimization TMLR
Nobari, Regenwetter, Giannone, Ahmed *2025*

Reparameterized Multi-Resolution Convolutions for Long Sequence Modelling NeurIPS
Cunningham, Giannone, Zhang, Deisenroth *2024*

Constraining Generative Models for Engineering Design with Negative Data TMLR
Regenwetter, Giannone, Srivastava, Gutfreund, Ahmed *2024*

Aligning Optimization Trajectories with Diffusion Models NeurIPS
Giannone, Srivastava, Winther, Ahmed *2023*

Diffusing the Optimal Topology: A Generative Optimization Perspective IDETC23
Giannone, Ahmed *2023*

Unifying Molecular and Textual Representations via Multi-task LM ICML
Christofidellis*, Giannone*, Born, Winther, Laino, Manica *2023*

Accelerating Material Design with GT4SD Nature npj Computational Materials
GT4SD Team (Core Contributor) *2023*

Few-Shot Diffusion Models SBM@NeurIPS
Giannone, Nielsen, Winther *2022*

SCHA-VAE: Hierarchical Context Aggregation for Few-Shot Generation ICML
Giannone, Winther *2022*

Method and apparatus for semantic segmentation and depth completion US Patent (NAVER)
Chidlovskii, Giannone *2022*

JM1: Worst-group Generalization by Group Interpolation NeurIPS-W
Giannone, Havrylov, Massiah, Yilmaz, Jiao *2021*

Hierarchical Few-Shot Generative Models NeurIPS-W
Giannone, Winther *2021*

Transformation-aware Variational Autoencoders Technical Report
Giannone, Saremi, Masci, Osendorfer *2020*

Input-filtering NeuralODEs for spiking data NeurIPS-W
Giannone, Anoosheh, Quaglino, D'Oro, Masci, Gallieri *2020*

$\mathcal{T}$-VAE: No Representation without Transformation NeurIPS-W
Giannone, Masci, Osendorfer *2019*

Learning Common Representation from RGB and Depth Images CVPR-W
Giannone, Chidlovskii *2019*

Projects & Open Source

its-hub: A Python library for inference-time scaling 2025

- Contributor.
- Inference-Time Scaling for Language Models.
- Focus on Mathematical Reasoning.
- Contributed Entropic Particle Filtering algorithms and new benchmark.

Text2CAD: Democratizing Engineering Design. Prompt by Prompt. 2023

- Co-Lead.
- DesignX. Team of engineers and researchers based at MIT and Caltech.
- Generative tool that allows users to create CAD models using natural language prompts.
- The tool is designed to be user-friendly and accessible to non-experts, enabling a wide range of users to quickly create complex CAD models without the need for specialized training.

GT4SD: Generative Toolkit for Scientific Discovery 2022

- Core Contributor.
- Library leveraging conditional generative models for accelerated discovery.
- Work on Diffusion Models for images and 3D molecule conformation. The GFlowNet framework. Property Prediction module. Public Hub for model upload. Training Pipelines. Documentation. Tutorials. Testing. CI/CD. Server and Client API. Docker Images for CPU and GPU.

Grants & Awards

GPU Grant, LUMI-G, EuroHPC Copenhagen, Denmark
Nov 2023

PI, Efficient Pre-training of Large Generative Models for Constrained Design

Grant, Otto Møensted's Foundation Copenhagen, Denmark
Dec 2022

Research Grant

Grant, Independent Research Fund Denmark Lyngby, Denmark
Jun 2020

DFF PhD Grant

Grant, Perception as Generative Reasoning NeurIPS 2019
Oct 2019

Awarded Complimentary Conference Registration by DeepMind

Grant, Pi School Rome, Italy
Oct 2018

Full Tuition for the School of AI (3% acceptance rate)

Certificate of Award, Tsinghua University Beijing, China
Aug 2018

Prize for Outstanding Accomplishments, Deep Learning Summer School

1st Pick, Excellence Path, Master's Degree, Data Science Rome, Italy
Mar 2018

Admission based on the First year's Academic Performance

Participation in the School for Advanced Studies

Academic Service

Reviewer

Conference: ICML19, ICCV19, AAAI20, ICML21 (top 10%), AISTATS21, ICML22, NeurIPS22, CVPR23, NeurIPS23, ICML24, ICLR25, CVPR25, NeurIPS25, ICLR26, ICML26, NeurIPS26, EMNLP26

Journal: TPAMI, TMLR

Workshop: NeurIPS-IBW20, NeurIPS-MetaLearn21, ICML-DeployableGenAI23, ACL-LanguageMolecules24

Teaching

Teaching: Deep Learning (DTU 02456), Bayesian Machine Learning (DTU 02477), Advanced Machine Learning (DTU 02460)

Supervision: two special courses (9 months), two master's thesis (6+6 months), 18 final projects

Talks

Algorithmic Methods for Data Mining (Sapienza University), Bayesian Reading Group (DTU), MLLS Center (KU), UCL-NLP (UCL), Amazon Alexa (Cambridge), DeCoDE Lab (MIT)

Volunteering

PAISS18, NeurIPS18, ELLIS Unit Copenhagen, MLLS, LeadTheFuture

Skills

Languages

- Python (proficient); R, Matlab (good knowledge); C, Java, JavaScript (basic knowledge)

Research

- Accelerate, HF Transformers, LaTeX, NLTK, OpenCV, PyTorch, SpaCy, TensorFlow, trl, verl

Software

- AWS, CVX, Docker/podman, FastAPI, Gradio, Linux, MinIO, MongoDB, MySQL, Travis, vLLM, LangGraph, LangFlow, Langfuse, OpenRouter